LESSON 6.2

Division Expressions with Fractions and Mixed Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing a fraction by a fraction means, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Rivera is dividing a 3/4-pound evidence bag into portions that each weigh 1/8 pound. She needs to know exactly how many portions she can make for testing. Can you solve the case?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Dividend *Spanish: Dividendo*	The total number being divided into equal groups (the number inside the division bracket).	In 8/9 ÷ 2/9, 8/9 is the dividend.
Divisor *Spanish: Divisor*	The number of equal groups you are dividing into (the number outside the division bracket).	In 8/9 ÷ 2/9, 2/9 is the divisor.
Reciprocal *Spanish: Recíproco*	A fraction turned upside down.	2/9 → 9/2
Quotient *Spanish: Cociente*	The answer when you divide.	8/9 ÷ 2/9 = 4
Simplify *Spanish: Simplificar*	To write a fraction with smaller numbers that names the same amount, like 2/4 = 1/2.	8/12 → GCF is 4 → 8÷4 / 12÷4 = 2/3
Mixed number *Spanish: Número mixto*	A whole number and a fraction written together, like 2 1/2. It must be rewritten as an improper fraction before dividing.	2 1/2 = 5/2

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A pitcher has 2/3 liter of juice. Each cup holds 1/6 liter. How many cups can be filled?

1. Write the division: 2/3 ÷ 1/6.
2. Flip the divisor and multiply: 2/3 × 6/1 = 12/3.
3. Simplify: 12/3 = 4. Check: 4 × 1/6 = 4/6 = 2/3.

Answer: 4 cups

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's try 1/2 ÷ 1/4. How many 1/4s fit in 1/2? Rename 1/2 as 2/4.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- To divide by _____ , I multiply by its reciprocal _____ .
- The quotient counts how many groups of _____ fit inside _____ .
- I did not need a common denominator because _____ .
- After multiplying across, I simplified _____ to _____ .

Word bank reciprocal | dividend | divisor | quotient | keep | change | flip | multiply across | simplify | groups | fit inside | tape diagram

WATCH OUT!

A common mistake

A common mistake in Divide Fractions is flipping the wrong fraction: students flip the dividend (the first fraction) instead of the divisor (the second fraction). For example, solving 3/4 ÷ 5/8 by flipping 3/4 to 4/3 and keeping 5/8 gives 4/3 × 5/8 = 20/24 = 5/6

LESSON 6.2

Division Expressions with Fractions and Mixed Numbers

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 1/2 ÷ 1/4?

(A) 1/8 (B) 1/2
(C) 2 (D) 4

STEPS TO TRY

1. This asks how many 1/4-size pieces fit inside 1/2 — the pieces are smaller than the amount.
2. Multiply by the reciprocal: keep 1/2, change ÷ to ×, and flip the divisor 1/4.
3. 1/4 flips to 4/1, so multiply 1/2 × 4/1 across the tops and bottoms, then simplify your fraction.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is 3/4 ÷ 1/8?

(A) 3/32
(B) 6
(C) 8/3
(D) 24

STEPS TO TRY

1. How many 1/8-pieces fit into 3/4? Eighths are small, so expect several.
2. Keep 3/4, change ÷ to ×, and flip 1/8 to its reciprocal.
3. Set up 3/4 × 8/1, multiply tops and bottoms, then simplify the product.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the reciprocal of 3/5?

(A) 1/3
(B) 3/5
(C) 5/3
(D) 5/5

STEPS TO TRY

1. Re-read 'reciprocal' — it's about the two numbers of the fraction trading places.
2. Swap the numerator (top number) with the denominator (bottom number). Nothing else changes.
3. In 3/5, the 3 is on top and the 5 is on the bottom. Trade their spots and match the result to a choice.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

4 **Circle the letter of the best answer. Show how you know.**

What is the first step in evaluating 2 1/2 ÷ 1/4?

(A) Rewrite 2 1/2 as the improper fraction 5/2

(B) Flip 2 1/2 to 1/2 2

(C) Divide 2 by 1/4 and ignore the half

(D) Add 2 + 1/2 + 1/4

STEPS TO TRY

1. Which number in this problem is a mixed number?
2. How many halves are in 2 1/2?
3. Rewrite it as one fraction before multiplying by the reciprocal.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does 5/6 ÷ 1/12 = 10? How does the bar model show this?

SENTENCE STARTER 5/6 ÷ 1/12 = 10 because 5/6 is the same as ___ twelfths, and ___ twelfths divided by 1 twelfth = ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help